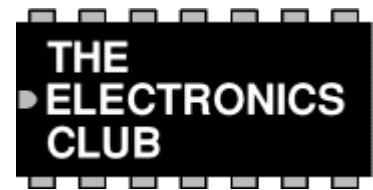


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Network Lead Tester Project



A kit for this project is available from [RSH Electronics](#).

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This project was designed for confirming the continuity and correct wiring of **computer network leads** which have 8 wires, but it can be used to check any lead with up to 8 wires by using appropriate connectors. For example stereo audio leads with 5-pin DIN plugs can be checked by just using the first 5 LEDs. The tester is simple but it can save a great deal of time when making up leads and it is much cheaper than the more sophisticated alternatives!

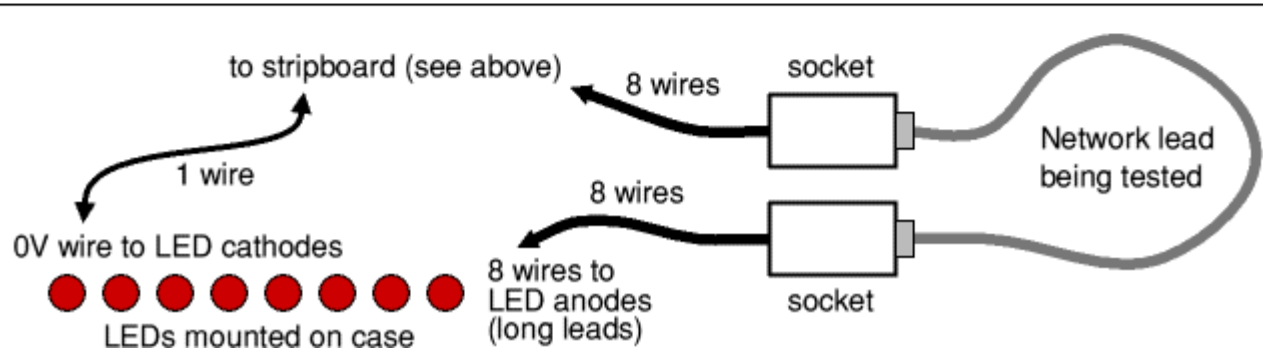
The tester works by connecting each wire to an output at one end and an LED at the other end. The outputs are switched on one at a time in sequence so that a correctly wired lead will make each LED light in turn. The 4017 chip counts up to 10 so there is a pause (for the 9th and 10th counts) before the LED sequence repeats. If the LEDs light up in the wrong sequence one (or both) of the connectors is wired wrongly. If an LED fails to light it indicates a broken connection. Please note that the RJ45 computer network plugs cannot normally be re-wired, instead they must be cut off and replaced.

This project uses a [555 astable](#) circuit to provide the clock pulses for the [4017 counter](#).

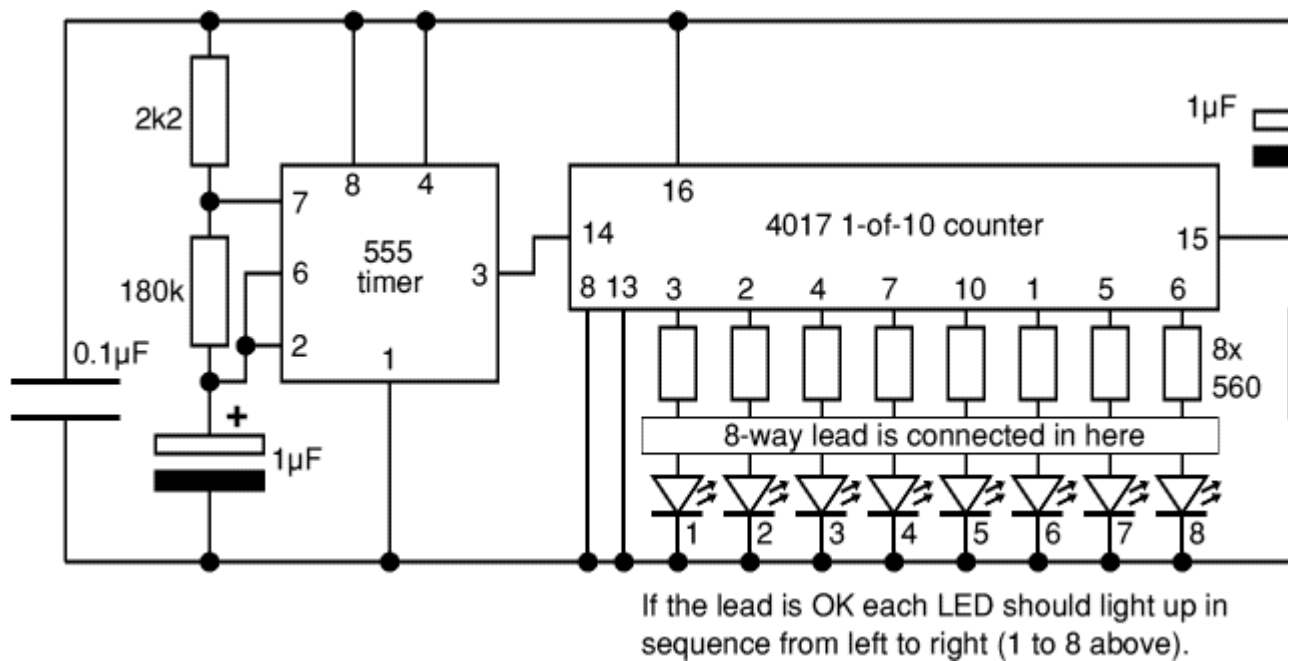
Parts Required

- resistors: 2.2k, 22k, 180k, 560 ×8
 - capacitors: 0.1μF, 1μF 16V radial ×2
 - red LEDs ×8
 - 555 timer IC
 - 4017 counter IC
 - DIL sockets for ICs: 8-pin, 16-pin
 - push switch
 - battery clip for 9V PP3
 - RJ45 computer network sockets ×2 (a dual socket for example)
(or suitable connectors for the leads you wish to test)
 - stripboard: 16 rows × 19 holes
-

Stripboard Layout



Circuit diagram



A kit for this project is available from [RSH Electronics](#). If you are new to electronics buying a kit is a good way to be sure you have the correct parts for the project.



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